MODULE 47 · DOMAIN 5 - OPERATIONAL EXCELLENCE AND AUTOMATION

AI/ML Observability and Model Drift

A model can be up, fast and wrong. Watch its inputs, its outputs, its fairness and its cost, and trace every step an agent takes.

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A model can be **available, fast and wrong**. Its quality has to be measured, because no error code will report it.

SageMaker Model Monitor

MONITOR	DETECTS	NEEDS
Data quality	Inputs drifting from training data	Baseline, data capture
Model quality	Accuracy, precision falling	Ground truth labels
Bias drift	Predictions becoming unfair	SageMaker Clarify config
Feature attribution drift	Different features driving results	SageMaker Clarify baseline

ON THE EXAM

Inputs only: data quality. Correctness: model quality, which needs labels.

How a drift monitor works

01
Baseline
Statistics from training data

02
Capture
Endpoint records requests and responses

03
Schedule
Compare capture with baseline

04
Alarm
CloudWatch, then retrain pipeline

ON THE EXAM

No baseline, no drift: the comparison needs a reference.

Observing Amazon Bedrock

SIGNAL	SOURCE
Invocations, latency, throttles	CloudWatch runtime metrics
Input and output tokens	CloudWatch metrics, per model
Full prompts and responses	Model invocation logging, off by default
Cost per application	Application inference profiles with tags
Guardrail interventions	Guardrail metrics and traces

ON THE EXAM

Token metrics show how much; invocation logs show which prompts.

Tracing agents

NEED	FEATURE
Every step of an agent run	CloudWatch generative AI observability
Model calls, tools, memory, gateway	AgentCore observability spans
Strands, LangChain, LangGraph	OpenTelemetry instrumentation
Tokens, latency, errors per agent	Curated dashboards

ON THE EXAM

Why did the agent do that? End-to-end trace of the run, not just the model call.

Drift in predictive models and in LLM apps

PREDICTIVE MODELS

Model Monitor

- Tabular inputs and labels
- Statistical baseline
- Ground truth for accuracy
- Retrain on drift

LLM APPLICATIONS

Evaluation on live samples

- Free-text inputs and answers
- Judge model scores
- Sample from invocation logs
- Fix prompt, model or knowledge base

THE TIEBREAKER

LLM answers getting worse with no labels: score sampled production responses with an evaluation job.

Reading the scenario

WHEN THE QUESTION SAYS	IT IS TESTING
Inputs no longer look like training data	Data quality monitor
Accuracy falling, labels arrive later	Model quality monitor
Predictions becoming unfair to a group	Bias drift with Clarify
Which application drives token spend	Invocation logs, inference profiles
Why did the agent take that path?	Generative AI observability traces
LLM answers worse, no labels	Evaluation on sampled traffic

ON THE EXAM

Inputs, correctness, fairness, cost, and the agent's path.

> Pellham Bank's fraud model runs on a SageMaker AI endpoint. New merchant categories launched, and transaction features now look quite different from the training data, but nothing alerted.
>
> The team wants an alert when production inputs drift from the training data.

What should the solutions architect implement?

A. A Model Monitor data quality schedule, using a baseline from the training data and endpoint data capture, with a CloudWatch alarm

B. A CloudWatch alarm on endpoint latency

C. Retrain the model every week regardless

D. A Clarify bias report at deployment

◆ PAUSE AND DECIDE

What should the solutions architect implement?

A Data quality monitor
Compares live inputs with the training baseline.

B Latency alarm
Drifted inputs are served just as fast.

C Weekly retraining
Costly, blind, and still no alert.

D Bias report once
Fairness, at one moment, not input drift.

THE SIGNAL

Production inputs drifting from training data: data quality monitor.

> Pellham Bank also needs to know whether the fraud model's precision and recall are falling. Investigators confirm whether a transaction was fraud about two weeks after it happens.
>
> The input distribution is already monitored.

What should the solutions architect add?

A A second data quality monitor

B An A/B test against the previous model

C Alarms on invocation errors

D A Model Monitor model quality schedule that merges the investigators' labels with captured predictions

◆ PAUSE AND DECIDE

What should the solutions architect add?

A Data quality again

Inputs only; says nothing about correctness.

B A/B test

Compares models, does not monitor one.

C Error alarms

Wrong predictions are not errors.

D Model quality monitor

Labels plus predictions give precision and recall.

THE SIGNAL

Is the model still correct, with labels arriving later: model quality monitor.

> Pellham Bank's credit model passed a fairness review before launch. The regulator now requires ongoing evidence that approval rates do not drift against applicants over 60.
>
> Results must be recorded on a schedule.

What should the solutions architect implement?

A. A data quality monitor on the age feature

B. Remove age from the model inputs

C. A Model Monitor bias drift schedule using SageMaker Clarify, with age as the facet

D. A quarterly manual review

◆ PAUSE AND DECIDE

What should the solutions architect implement?

A Data quality on age

Watches the age distribution, not outcomes.

B Remove age

Proxies can still carry bias; no evidence produced.

C Bias drift monitor

Scheduled Clarify metrics for the age group.

D Manual review

Infrequent, not continuous.

THE SIGNAL

Ongoing evidence that outcomes stay fair to a group: bias drift monitor.

> Pellham Bank runs four applications on the same Bedrock model. Spend on the model doubled last month, and finance wants to know which application caused it and which prompts are largest.
>
> Only aggregate CloudWatch token metrics are available today.

What should the solutions architect do?

A Read CloudTrail management events

B Give each application its own tagged application inference profile, and enable model invocation logging to analyse prompt sizes

C Enable X-Ray tracing on the model

D Divide the bill evenly between the applications

◆ PAUSE AND DECIDE

What should the solutions architect do?

A CloudTrail
No token counts or prompt contents.

B Profiles and invocation logs
Cost per application; prompts per request.

C X-Ray
Timing, not token attribution.

D Even split
Guesswork.

THE SIGNAL

Which application and which prompts drive token spend: inference profiles and invocation logs.

> Pellham Bank's customer service agent, built with Strands Agents on Bedrock AgentCore, sometimes calls the wrong tool and takes 40 seconds to answer.
>
> Engineers need to see each step of a run: model calls, tool calls, memory reads, with timings.

What should the solutions architect use?

A Model invocation logs alone

B CloudTrail data events

C An Application Signals availability SLO

D CloudWatch generative AI observability, with the agent instrumented through OpenTelemetry

◆ PAUSE AND DECIDE

What should the solutions architect use?

A Invocation logs

Model calls only; no tools, memory or timeline.

B CloudTrail

API audit, not a run trace.

C SLO

Tells you it is slow, not why.

D Generative AI observability

End-to-end trace of every step, with timings.

THE SIGNAL

Every step of an agent run, with timings: generative AI observability traces.

> Pellham Bank's policy assistant answers staff questions from a knowledge base that changes weekly. Staff say answers have become vaguer over three months. There are no labels for production questions.
>
> The team wants continuing measurement of answer quality in production.

What should the solutions architect implement?

A. Sample responses from invocation logs each day and score them with a Bedrock LLM-as-a-judge evaluation, publishing scores to CloudWatch with an alarm

B. A Model Monitor data quality schedule

C. A stricter Bedrock guardrail

D. An alarm on output token count

◆ PAUSE AND DECIDE

What should the solutions architect implement?

A Judge on live samples

Measures quality without labels, over time.

B Model Monitor

Built for structured inputs and SageMaker endpoints.

C Stricter guardrail

Blocks content; does not measure quality.

D Token alarm

Length is not quality.

THE SIGNAL

Generative answers drifting, no labels: evaluate sampled production responses.

What to carry out of this module

- **Model Monitor**: data quality, model quality, bias, attribution.
- **Baseline, capture, schedule, alarm**, then retrain.
- **Bedrock**: token metrics, invocation logs, inference profiles.
- **Generative AI observability** traces every agent step.
- **LLM drift**: judge-scored samples from production.

> ON THE EXAM
>
> Inputs, correctness, fairness, cost, and the agent's path.